

ECE276 Project3

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I. INTRODUCTION

Visual-Inertial SLAM (Simultaneous Localization and Mapping) aims to estimate the trajectory of a robot while simultaneously reconstructing the surrounding environment. This problem is fundamental in robotics, enabling applications such as autonomous navigation and mapping.

In this project, we implement an Extended Kalman Filter (EKF)-based visual-inertial SLAM system using IMU measurements and stereo visual features. The pipeline consists of IMU-based motion prediction, landmark mapping through visual observations, and full SLAM integration to refine trajectory estimation.

II. PROBLEM FORMULATION

We aim to estimate:

- Robot pose: $T_t \in SE(3)$
- Landmark positions: $m \in \mathbb{R}^{3 \times M}$

Inputs:

- IMU measurements: linear velocity v_t and angular velocity ω_t
- Stereo feature observations: z_t

Goal:

- Estimate trajectory over time
- Recover 3D landmark positions

III. TECHNICAL APPROACH

A. Overview

This project implements a visual-inertial SLAM system using an Extended Kalman Filter (EKF). The pipeline consists of three main components:

- IMU-based pose prediction (Part 1),
- Landmark mapping using stereo observations (Part 3),
- Full SLAM with joint state estimation (Part 4).

The EKF framework allows us to fuse high-frequency IMU measurements with lower-frequency but geometrically informative visual observations.

B. Part 1: IMU Localization (Prediction)

The robot pose evolves on the Lie group $SE(3)$, which represents rigid body transformations.

1) *Motion Model:* Given IMU measurements:

- linear velocity $v_t \in \mathbb{R}^3$
- angular velocity $\omega_t \in \mathbb{R}^3$

we construct the twist vector:

$$\xi_t = \begin{bmatrix} v_t \\ \omega_t \end{bmatrix} \quad (1)$$

and its corresponding Lie algebra representation $\hat{\xi}_t \in \mathfrak{se}(3)$.

The pose is propagated using the exponential map:

$$T_{t+1} = T_t \cdot \exp(\hat{\xi}_t \Delta t) \quad (2)$$

This formulation ensures that the pose remains on $SE(3)$.

2) *EKF Prediction Step:* The EKF prediction updates the state mean and covariance:

$$\mu_{t+1} = f(\mu_t, u_t) \quad (3)$$

$$\Sigma_{t+1} = F_t \Sigma_t F_t^T + W_t Q_t W_t^T \quad (4)$$

where:

- F_t is the Jacobian of the motion model,
- Q_t is the IMU noise covariance,
- W_t maps process noise into the state space.

Practical insight: Since the IMU measurements are integrated over time, small noise in velocity and angular rate accumulates, leading to drift in long trajectories. This is clearly observed in the Part 1 results.

C. Part 3: Landmark Mapping

In this part, the robot trajectory is assumed known, and the goal is to estimate landmark positions.

1) *Stereo Triangulation:* Each landmark is observed in both left and right cameras. Given pixel coordinates:

$$z = [u_L \quad v_L \quad u_R \quad v_R]^T \quad (5)$$

depth is computed from disparity:

$$d = u_L - u_R \quad (6)$$

and the 3D point is reconstructed using camera intrinsics and stereo baseline:

$$Z = \frac{f \cdot b}{d}, \quad (7)$$

$$X = \frac{(u_L - c_u)Z}{f}, \quad (8)$$

$$Y = \frac{(v_L - c_v)Z}{f} \quad (9)$$

The point is then transformed into the world frame using the known pose.

2) *Observation Model:* The measurement model projects a 3D landmark into the stereo image:

$$z = h(T_t, m_j) \quad (10)$$

The Jacobian H is computed with respect to the landmark state.

3) *EKF Update*: Using your formulation:

$$K_t = \Sigma_0 H^T (H \Sigma_0 H^T + V)^{-1} \quad (11)$$

$$\mu = \mu_0 + K_t(z - h(\mu_0)) \quad (12)$$

$$\Sigma = (I - K_t H) \Sigma_0 \quad (13)$$

Implementation details:

- Only landmarks with valid disparity are initialized.
- Landmarks are updated only when observed.
- Invalid measurements (e.g., -1 entries) are skipped.

Key challenge: Depth estimation becomes unstable when disparity is small, leading to noisy landmark initialization.

D. Part 4: Visual-Inertial SLAM

In the final part, both robot pose and landmark positions are estimated jointly.

1) *State Representation*: The combined state includes:

$$x = \begin{bmatrix} x_{\text{pose}} \\ x_{\text{landmarks}} \end{bmatrix} \quad (14)$$

2) *SLAM Pipeline*: At each timestep:

- 1) Predict pose using IMU (same as Part 1)
- 2) Select visible landmarks
- 3) Project landmarks into image plane
- 4) Compute innovation:

$$y = z - h(x) \quad (15)$$

5) Update state using EKF

3) *Drift Correction Mechanism*: The key advantage of SLAM comes from the correction step:

- IMU provides short-term motion accuracy
- Visual landmarks provide global geometric constraints

This allows previously accumulated errors to be corrected when landmarks are re-observed.

4) *Noise Tuning*: Performance depends heavily on noise parameters:

- High IMU noise $\rightarrow$ more reliance on vision
- High vision noise $\rightarrow$ smoother but less correction

In practice, a balance is required to avoid:

- overconfidence in IMU (drift)
- overreaction to noisy features (instability)

5) *Computational Considerations*: Since the number of landmarks can be large:

- Only a subset of landmarks is updated per step
- Sparse updates are used to reduce computation

This improves efficiency while maintaining accuracy.

Summary: The EKF-based SLAM framework successfully integrates motion and observation, significantly improving trajectory estimation compared to IMU-only localization.

IV. RESULTS

A. Part 1: IMU Trajectory

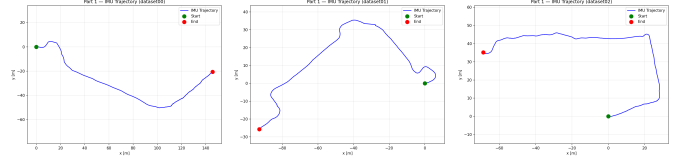


Fig. 1: IMU-only trajectory estimation.

Observation: IMU-only trajectory suffers from significant drift due to integration of noisy measurements.

B. Part 3: Landmark Mapping

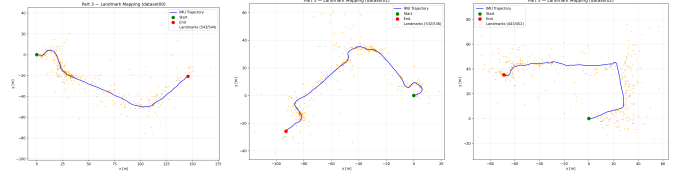


Fig. 2: Landmark mapping results.

Observation: Landmarks are distributed around the trajectory and generally align with the robot motion. Some outliers exist due to noise and imperfect feature tracking.

C. Part 4: Visual-Inertial SLAM

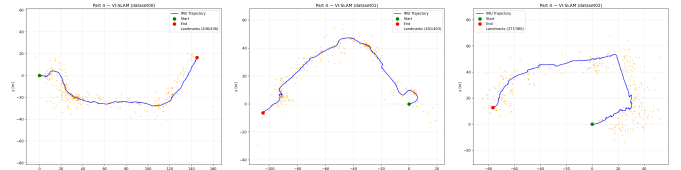


Fig. 3: Final VI-SLAM trajectory with landmarks.

Observation: Compared to Part 1, the trajectory is significantly improved. Visual updates correct drift and produce a more consistent path.

D. Video Results

In addition to the static plots, we also generated trajectory visualization videos for all three datasets:

- trajectory_video_dataset00.mp4
- trajectory_video_dataset01.mp4
- trajectory_video_dataset02.mp4

These videos provide qualitative validation of the SLAM performance and are included in the submission.

V. DISCUSSION

What works:

- EKF successfully integrates IMU and visual information
- Landmark updates reduce drift significantly

Limitations:

- Landmark depth estimation is noisy (expected due to limited motion in z-axis)
- Some outliers remain due to feature tracking errors

Future improvements:

- Better feature filtering (e.g., RANSAC)
- More robust data association
- Bundle adjustment refinement

VI. CONCLUSION

In this project, we implemented a complete visual-inertial SLAM system using EKF. The combination of IMU prediction and visual updates effectively reduces drift and improves trajectory estimation. The results demonstrate the importance of sensor fusion in robotics applications.